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EDITORIAL

From the Editor's Desk: A New Beginning in the Journey of Medical Science

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It gives me immense pleasure and a deep sense of responsibility to welcome you to the inaugural issue of the JMN Journal of Medical Sciences, the official academic publication of JMN Medical College.

The birth of a medical journal is more than the beginning of another publication. It represents the collective aspiration of an institution to create, share, and advance knowledge. It is a commitment to scientific enquiry, academic excellence, ethical practice, and the continuous pursuit of better healthcare.

Medicine is an ever-evolving discipline. Every day brings new discoveries, changing treatment paradigms, technological advances, and fresh perspectives on patient care. In this rapidly changing environment, the ability to critically evaluate evidence and contribute meaningful knowledge is as important as the ability to apply existing knowledge. We hope that this journal will provide a platform where such ideas can be expressed, discussed, challenged, and translated into clinical practice.

Our vision is to develop this journal as a forum for original research, clinical studies, review articles, case reports, perspectives, and scholarly discussions across the diverse fields of medicine and allied health sciences. We particularly encourage young faculty, undergraduate students, and researchers to come forward and share their work. A journal should not merely record established knowledge; it should also encourage the curiosity and creativity of the next generation of healthcare professionals.

At the same time, scientific publication carries a profound responsibility. The credibility of a journal depends upon the quality, originality, transparency, and ethical integrity of the work it publishes. We are therefore committed to maintaining rigorous peer review, appropriate research ethics, responsible authorship, transparency in reporting, and high editorial standards. We aspire to build a publication culture in which quality is valued over quantity, and integrity over recognition.

This journal also seeks to bridge the gap between research and real-world patient care. The ultimate purpose of medical knowledge is not simply to be published, but to improve understanding, enhance clinical decision-making, and ultimately benefit our patients and communities. We hope that the contributions to this journal will therefore have relevance not only within academic institutions but also in everyday healthcare practice.

The inaugural issue is only the first step of a much longer journey. The success and growth of this journal will depend upon the collective support of our faculty, researchers, students, reviewers, editorial board members, and contributors. I invite each member of our academic community to consider this journal as your platform for sharing knowledge and your contribution to the advancement of medicine.

I would like to express my sincere gratitude to the leadership of JMN Medical College, the members of the editorial and advisory boards, our reviewers, authors, teachers, students, and

everyone who has contributed to bringing this vision to life.

May this journal grow from a new publication into a respected platform for scholarship, collaboration, and innovation.

With great enthusiasm and optimism, I welcome you to the first issue of the JMN Journal of Medical Sciences.

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Original Article

A Study on Urinary Incontinence and Its Risk Factors Among Post-menopausal Women in a Rural Area of West Bengal

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Abstract

Background: Urinary incontinence (UI) is the involuntary loss of urine which is objectively demonstrable and is a social and hygienic problem. UI is a chronic debilitating disease which is often underreported, but has significant impact on daily activities, and is thus of public health importance. **Objectives:** To find out the proportion of rural post-menopausal women having UI and its associated risk factors. **Methods:** A cross-sectional study was conducted in a rural area of Barasat among 120 elderly women ($\geq$50 years) selected by cluster sampling (8 subjects each from 15 clusters) with a pre-designed questionnaire. Chi-square test and logistic regression were applied to find associations between variables. $P<0.05$ was considered statistically significant. Data were analyzed with SPSS version 16.0. **Results:** 43 (35.8%) of 120 women were found to have UI. The most prevalent type was stress UI (18.3%), followed by mixed UI (10.8%) and urge UI (5.8%). On multivariable analysis, illiteracy (AOR 2.41 [1.02–5.69]), NVDs >3 (AOR 3.37 [1.54–7.37]), history of gynaecological operation (AOR 3.84 [1.16–12.66]), chronic cough (AOR 2.69 [1.21–5.99]), and LUTS (AOR 2.63 [1.15–6.00]) remained significantly associated with UI. Only 25 (20.8%) sought medical help. **Conclusion:** Rural women are at high risk of developing UI, and most affected women did not seek treatment, which is a matter of concern. Awareness generation regarding UI may help improve health-seeking behaviour.

Key Words: Lower urinary tract symptoms, Post-menopausal women, Risk factors, Rural health, Urinary incontinence

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I. Introduction

Urinary incontinence is a common distressing problem amongst adults living in the community [1]. Control of the micturition reflex is a complex physiological and anatomical process that may often fail, particularly in women. According to the International Continence Society (ICS), urinary incontinence (UI) is defined as: the complaint of any involuntary loss of urine that is a social or hygienic problem [2]. Worldwide, it has been observed that over 200 million people are living with urinary incontinence that may hamper their normal physical activity [3]. It is three times more common in women than men [4]. Urinary incontinence is a troublesome and probably underreported disorder, especially in women [5]. An estimated 19% of women under the age of 45 experience UI and 29% of women over the age of 80 also experience this condition [6] . In general, the overall prevalence of UI increases with age

[7]. Numerous epidemiologic studies show that the incidence of UI increases with age, with the range of prevalence estimates among community dwelling patients varying enormously from 2% to 58%[8]. Postmenopausal or elderly women are the most affected with a mean prevalence of 34% [9]. Women suffering from urinary incontinence constitute about 70-80% of female patients reporting to a physician due to urinary system diseases [10]. It is estimated that women of different ages have involuntary urine loss and specially 9-39% of women over 60 years report daily urinary leakage [11,12]. The prevalence of urinary incontinence is gradually increasing and may diminish quality of life and disrupt their daily routines [13]. Recently, as the average life expectancy of women has increased and an increasing number of women maintain various social roles, urinary incontinence (UI) has become an important medical and social issue [14]. Urinary continence is due to the fact that bladder pressure remains lower than urethral closure pressure, and urinary incontinence may result from bladder or urethral impairment; when closure pressure is lower than bladder pressure, urinary leakage occurs [15] . Among elderly women, many are in the postmenopausal phase; so the low estrogen production produces symptoms from the autonomic nervous system such as flushing and perspiration and psychiatric symptoms such as depression. Furthermore, the mucous membranes of the genitourinary tract among the elderly women undergo atrophy resulting in vaginitis, dyspareunia and cysto-urethritis [16]. Basically there

are three types of incontinence : stress urinary incontinence (SUI); urge urinary incontinence (UUI), and mixed urinary incontinence (MUI), which associates the first two [17] . Stress is most common occurring in pure or mixed forms, in 80% women with incontinence defined as the complaint of involuntary leakage on effort or exertion, or on sneezing or coughing. [11,14,16]

. Urinary incontinence hampers the person's freedom and ability to function in everyday life. It has therefore become a public health problem, with a high economic and human impact [20]. Although the interest is growing regarding the investigation, treatment, and management of incontinence, its prevalence in the general population has so far been truly based on some estimates made upon selected groups of people of different ages [21]. Despite its high prevalence, very little is known about risk factors associated with UI, and the etiology of urinary incontinence is poorly noted. Various factors which are associated with lifestyle are thought to aggravate lower urinary tract symptoms and results in urinary incontinence, for example, ageing, pregnancy, vaginal delivery, obesity physical forces (exercise, work), smoking, caffeine and fluid intake, constipation, posture [15,18,19] . Therefore, this study was carried out to estimate the prevalence of Urinary Incontinence among the postmenopausal (>60 years) women and to find out the risk factors among them associated with Urinary Incontinence. Most studies on urinary incontinence have been carried out in developed countries and affluent study populations. Scarce data exist on its prevalence in India, specially in rural women, or in those with lower socioeconomic status. The condition is usually under reported as most of the women hesitate to seek help or report symptoms to medical practitioners due to the embarrassing and culturally sensitive aspect of this condition. So if we can achieve

to get an updated picture of UI among rural women, that will be of great importance in helping formulating strategies of prevention and control for UI and that further helps to reduce the disease burden in India . In this aspect this study examined prevalence of UI among rural women, and the risk factors associated with it among the study participants.

II. Materials and Methods

A community-based cross-sectional study was conducted from May to July 2022 in the rural area of Barasat, West Bengal comprising 64 villages. All the postmenopausal women(>=60 years) [22] in the study setting were included, but those who were critically ill patients or who were not willing to participate in the study or those with known psychiatric illness they were excluded from the study. Sample size will be calculated

using (Z2.P.Q/L2) as per cochran's formula. Where, P = Prevalence of cardiovascular risk, Z=

1.96 (95% confidence interval), L= absolute error of precision. Considering the following assumptions prevalence of UI among postmenopausal (>60 years) women was found to be 27.7% [23] among women in rural India from a similar study. Taking L = 10%, sample size(n) came= 75.71. Considering a design effect of 1.5 for cluster sampling, it came =(75.51 $\times$ 1.5)= 113 and taking 10% non-response rate, final sample size came 120. A two stage cluster sampling was done to select study participants. In study area there were 64 villages, from these villages at first stage 12 villages or cluster were selected randomly. With the help of local maps and local people after going to the centre of the first village, at first one direction was chosen randomly by rotating bottle head. Going to that direction one house was chosen. That was the first house to elderly women participant as per inclusion criteria. Next adjacent houses were visited until 10 participants were selected (At the second stage 10 elderly women were selected from each cluster (120/12=10). When there was end of road, next lane was taken to complete sample size. Then the same procedure was repeated in rest of all 11 clusters. [Fig 1]. Face to face interview of the participants were conducted with the pre-designed, pre-tested questionnaire and anthropometric and clinical examination was done. Urinary Incontinence, dependent variable was assessed by QUID Questionnaire [24]) was the dependent variable. Other independent variables were sociodemographic variable- Age in completed years, Level of Education (in completed years of Education), Religion, Caste, Family type, Socio economic class of the family (Modified B.G. Prasad Scale, 2020), Occupational Status ; reproductive variables -Menstrual cycle regularity, no of child birth, h/o abortion/PCOD, age of first child birth, age of marriage, marital status, complication during pregnancy, ever OCP usage; behavioural variables-Tobacco/Alcohol use, Physical Activity Status assessed by IPAQ- SF[25] (International Physical Activity Questionnaire- short form); Mental stress(assessed by PSS-4); others- H/O chronic illness, family h/o any chronic illness(Hypertension, Diabetes, Dyslipidemia), dietary Behaviour, BMI(body mass index). During data collection the past medical records of the participants were also assessed.

Ethical clearance was obtained. The study's purpose was described, and all participants gave their informed consent. Confidentiality and privacy of the participants were genuinely maintained throughout the whole study. Data were analyzed by Microsoft Excel 2019 and statistical package for social sciences (Version 16). Univariate logistic regression and multivariable logistic regression were done. P<0.05 was considered as statistically significant within 95% confidence interval.

III. Results

Majority of the study participants (44.2%) belonged to age group 70 years or above. The mean (standard deviation) of the age of the study participants was 60.88 ($\pm$7.65). Majority (95%) of the study participants were Hindu. About 59.2% participants were with no formal education. Most of the study participants (64.2%) belonged to others caste. Among the post menopausal age group women majority (61.6%) belonged to socio economic class 4.

About 78.3% were homemaker by occupation. Nearly 64.2% of post-menopausal age group women stay in the joint family. Nearly 32.5% of them were diabetic while 36.7% of them were hypertensive. [Table 1].

Nearly 32.5% of them were diabetic while 36.7% of them were hypertensive. Among the study participants 36.7% had chronic cough and 35% had h/o constipation. Nearly 22.5% of them had a history of gynecological operation. Among them 53.3% had more t han equals t o normal vaginal delivery (NVD), while 48.3% had a history of abortion . Around 25.8% had h/o fecal incontinence. About 37.5% of them were suffering from any one of the LUTSs. Among them 40% elderly women had low physical activity and 35% with high mental stress. According to BMI 25.8% were overweight and 2.5% were obese. [Table 2].

Forty-two (35%) out of 120 women were found having UI. The most prevalent type of UI was stress UI (18.3%);followed by mixed UI (10.8%) and urge UI (5.8%). It has been seen that 44.2% of post-menopausal age group aged 70years and above; 32.5% of aged 65- 69 years post-menopausal age group women and 23.3% post-menopausal age group women aged 60-64 years had urinary incontinence. Only 15 (30.6%) sought treatment for UI.

In bivariate logistic regression analysis, study participants, who were illiterate, having a history of gynecological operation, NVDs (>3), diabetic, hypertension, chronic cough, constipation, LUTS, had shown significantly greater odds of having UI. In multivariable model hypertension (adjusted odds ratio [AOR] - 2.15 [1.13-4.75]), chronic cough (AOR

- 4.50 [1.24-16.30]), and constipation (AOR - 8.58 [2.06-35.65]) remained significant adjusted with other significant variable in bivariate analysis [Table 3].

Table 1: *Distribution of study participants according to background characteristics (n=120)*

Parameters	Category	n (%)
Age (years)	60–64	28 (23.3)
	65–69	39 (32.5)
	≥70	53 (44.2)
Religion	Hindu	114 (95.0)
	Muslim	6 (5.0)
Educational level	Illiterate	71 (59.2)
	Below primary	14 (11.7)
	Primary	25 (20.8)
	Middle	6 (5.0)
	Secondary	4 (3.3)
Caste	General/Others	77 (64.2)
	SC	43 (35.8)
	OBC	5 (4.2)
Socioeconomic class	Class 1	2 (1.6)
	Class 2	5 (4.1)
	Class 3	17 (14.1)
	Class 4	74 (61.6)
	Class 5	22 (18.3)
Occupation	Homemaker	94 (78.3)
	Cultivator/Biri binder	26 (21.7)
Family type	Nuclear	43 (35.8)
	Joint	77 (64.2)

Table 2: *Distribution of study participants according to other characteristics (n=120)*

Parameters	Category	n (%)
Diabetes	Absent	81 (67.5)
	Present	39 (32.5)
Hypertension	Absent	76 (63.3)
	Present	44 (36.7)
Chronic cough	No	76 (63.3)
	Yes	44 (36.7)
Constipation	No	78 (65.0)
	Yes	42 (35.0)
NVD	≤3	64 (53.3)
	>3	56 (46.7)
LUTS	No	75 (62.5)
	Yes	45 (37.5)
H/O gynae. operation	No	93 (77.5)
	Yes	27 (22.5)
Faecal incontinence	No	89 (74.2)
	Yes	31 (25.8)
H/O abortion	No	58 (48.3)
	Yes	62 (51.7)
BMI	Underweight (<18.5)	2 (1.7)
	Normal (18.5–24.9)	84 (70.0)
	Overweight (25–29.9)	31 (25.8)
	Obese (≥30)	3 (2.5)
Physical activity	Low	48 (40.0)
	High	72 (60.0)
Mental stress	Low	78 (65.0)
	High	42 (35.0)

Table 3: *Factors associated with presence of urinary incontinence among study participants: logistic regression analysis (n=120)*

Variable	Category	Total n (%)	UOR (95% CI)	AOR (95% CI)
Diabetes	Absent	81 (67.5)	1 (Ref)	1 (Ref)
	Present	39 (32.5)	12.18 (4.93–30.09)	2.28 (0.56–9.26)
Hypertension	Absent	76 (63.3)	1 (Ref)	1 (Ref)
	Present	44 (36.7)	14.09 (5.67–34.99)	2.15 (1.13–4.75)
Chronic cough	No	76 (63.3)	1 (Ref)	1 (Ref)
	Yes	44 (36.7)	14.09 (5.67–34.99)	4.50 (1.24–16.30)
Constipation	No	78 (65.0)	1 (Ref)	1 (Ref)
	Yes	42 (35.0)	21.76 (8.23–57.31)	8.58 (2.06–35.65)
LUTS	No	75 (62.5)	1 (Ref)	1 (Ref)
	Yes	45 (37.5)	6.00 (2.63–13.65)	0.73 (0.17–3.01)
NVD	≤3	64 (53.3)	1 (Ref)	1 (Ref)
	>3	56 (46.7)	8.76 (3.62–21.20)	2.35 (0.62–8.83)
H/O gynae. op.	No	93 (77.5)	1 (Ref)	1 (Ref)
	Yes	27 (22.5)	4.62 (1.86–11.43)	0.52 (0.09–2.73)

UOR: Unadjusted odds ratio; AOR: Adjusted odds ratio; CI: Confidence interval

IV. Discussion

About 35% of the study population (postmenopausal age group) was found to have UI. The findings were consistent with findings of Prabhu and Shanbhag[26] (25.5%), Ansar et al.[27] (23.9%), and Seshan and Muliira[28] (33.8%), Abha et al.[29] (34.0%), Kılıç[30] (37.2%), and Sensoy et al.[31] (44.6%) have reported like this. Singh et al.[32] (21.8%), Ge et al.[33] (22.1%), Sumardi et al.[34] (13.0%), Abiola et al.[35] (12.6%), and Bodhare et al.[36] (10.0%) have reported less. This variance in the prevalence may be due to different study population, settings, and definition of UI used.

The study revealed that illiterate women are at more risk. Hence educational level plays a vital role in this regard. Sensoy et al.,[31] Seshan and Muliira,[28] and Ge et al.[33] had similar findings. A history of prolonged labor: it is an established cause of urinary tract injury-thus increases the risk of UI. The study had identified a history of prolonged labor as a risk factor for UI similarto Bodhare et al.[36] and Kılıç[30] with increase in number of childbirth chance of trauma to the urinary tract also increases. The study established NVD as a risk factorfor UI similar as Singh et al.,[32] Ge et al.,[33] Seshan and Muliira,[28] and Sensoy et al.[31]

A history of gynecological operation imposes risk of iatrogenic trauma to the urinary tract, thus increasing risk ofUI. The study revealed that also which is similar to findingsof Prabhu and Shanbhag,[26] Sensoy et al.,[31] Ge et al.,[33]and Singh et al.[32] Diabetes causes polyuria which laid additional burden on sphincters of urinary tract resulting in UI. The current study also establishes the fact same as Prabhu and Shanbhag,[26] Kılıç,[30] Ge et al.,[33] and Singh et al.[32] Constipation and chronic cough create additional stress on both the anal and urethral sphincter resultinginto UI. The studies by Prabhu and Shanbhag,[26] Bodhareet al.,[36] and Sensoy et al.[31] identified constipation and chronic cough both as risk factors of UI while Ge et al.[33]found constipation and Sumardi et al.[34] identified chroniccough as a risk factor. The study establishes LUTS as oneof the most important risk factors for UI similar to Sumardiet al.,[34] Sensoy et al.,[31] Ge et al.,[33] and Kılıç.[30]

The most prevalent type of UI was stress UI (18.3%), followed by mixed UI (10.8%) and urge UI (5.8%).This finding further corroborated earlier reports from epidemiological studies of UI in India in which stress incontinence was the most common type of UI among women with UI.[26,29,32] . Only 30.6% sought treatment for UI which is better compared to Prabhu and Shanbhag[26] (14.4%), Sarici et al.[37] (10.7%), Jokhio et al.[38] (15.7%) and worsecompared to Lasserre et al.[39] (39.7%).

V. Conclusion

Prevalence of UI is high in rural postmenopausal age group women. Most prevalent one is stress UI. Majority of them did not sought treatment for UI which is matter of concern. Generating awareness regarding UI may help to improve health-seeking behavior. In the current study, UI is predicted by certain modifiable risk factors, proper management of which is warranted to reduce the burden of UI and thus improving the health status of elderly women. Several treatment choices for UI are now available with greater effectiveness. For example, pelvic floor muscle exercise or perineal exercise is established to be effective first-line intervention to prevent the development of UI and further complication. Thus, encouraging pelvic floor muscle exercise or perineal exercise among the at risk women and those who are affected is useful in both the prevention and management of UI. An attempt

should be made by doctor or health personnel to screen at risk women for UI at every given opportunity by asking leading questions so that the affected can be timely referred for relevant management.

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Original Article

A Study on Assessment of Functional Status and Its Correlates Among the Geriatric Population in Rural India

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Abstract

Background: Functional ability is a proxy for healthy aging and is related not only to mental and physical health but also to social well-being. Identifying the determinants of functionality among the geriatric population can help evolve appropriate plans at both domiciliary and facility level. **Objectives:** To determine the magnitude of poor functional status among the geriatric population and its correlates. **Methods:** A community-based cross-sectional study was conducted among 246 geriatric people in a rural area of West Bengal from August to December 2022; data were analyzed using SPSS (version 16.0). Logistic regression was performed to identify correlates of poor functional status (activities of daily living [ADL] and instrumental ADL [IADL]). **Results:** 32.5% and 59.3% of participants were dependent for basic ADL and IADL, respectively. Age >70 years, female gender, less than primary education, widowed/separated status, joint family, poorest economic percentile, depression, and multimorbidity were associated with increased dependency. On multivariate analysis, age >70 years and depression remained significant for ADL; marital status, education, and family type additionally remained significant for IADL. **Conclusion:** Community-based comprehensive geriatric health and disability assessment should be provided, enabling early detection and delay of disabling disease progression, and supporting the well-being and independence of the elderly.

Key Words: Elderly, Functional status, Geriatric population, Predictors, Rural area

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I. Introduction

Aging is not uniformly experienced among aging adults. Some elderly people lead their life with a sense of fulfillment and satisfaction in their old age, while other older adults may become frail and experience declining physical abilities due to the decline of their physical

abilities and social significance.[1] The life expectancy has increased from 40 years in 1951-1967 years in 2015, a person has 25 years more to live than he would have 50 years back.[2] According to SRS 2015, the percentage of the geriatric population in the rural area is 8.3, and inthe urban area, it is 8.4.[3] Due to globalization, most of the families results in breaking up of traditional families and formation of nuclear families that results to make the elderly to live a life in their own without taking any support.

The loss of functional ability typically associated with aging is absolutely related to a person's chronological age. There is no such "typical" older person.[4] Healthy aging is more than just the absence of disease. For most of the geriatric population, the maintenance of functional ability has the highest importance.[4] Functional status is usually defined in terms of restrictions on the ability to function independently in terms of basic ADL (ability to perform tasks of daily living) and another type instrumental ADL (IADL).[5] Functional activity is the performance related to not only mental and physical health; but it may also determine the aspect of social well-being.[6]

Longevity actually results in chronic diseases which affect the functional ability that may compromise the ability to pursue the daily routine activities, forming a situation with a need for assistance. So reduction of the severe disability is one of the best key to holding down health and social costs as well as to improve the quality of life and maintain a healthy well- being.[4] In the developed world there are numerous studies on geriatric care in developed settings. However, evidence from rural India remains limited, highlighting the need for context-specific health-system planning and high-quality physical and mental health services. Detection of the determinants of functionality components among the geriatric population

will help to assist in evolving with appropriate plans at both domiciliary and facility level to eliminate their sufferings and disabilities and to maintain a proper livelihood.

With this background, an attempt was made to assessthe functional status and the factors associated with it, among the geriatric population in a rural area of Singur, West Bengal.

II. Materials and Methods

The study place was the service area of Rural Health Unit and Training Centre, Singur under Field practice area of All India Institute of Hygiene and Public Health (AIIH&PH, Kolkata comprising of 64 villages. It is located in Singur block of Chandannagore sub-division in Hooghly District, with a distance of 50 km from the state capital of Kolkata; well connected by train (on Howrah-Tarakeshwar railway line) and road (besides Durgapur expressway and NH2). The study area contains 64 villages from seven Gram Panchayats, namely, Singur-I, Nasibpur, Anandanagar, Bagdanga-Chhinamore, Boinchipota, and Baruipara-Paltagarh.

This is a community-based cross-sectional study conducted among the geriatric population (60 years and above) in the Rural service area of A I I H & P H , Rural Health Unit and Training Centre Singur, Kolkata, which consists of 64 villages, 15-cluster technique was implemented, i.e., 15 villages were to be selected from 64 villages by the first stage of the cluster sampling. 15 clusters were selected from 64 villages as per PPS method by cluster sampling. 246 participants were recruited from 6 clusters. Total of 64 villages from which 6 villages were randomly selected, during the period of 5 months from August to December 2022. Taking reference of a similar study, where is 20% prevalence (p) for impairment in activities of daily living (ADL), with an absolute precision (d) of 5% and 95% confidence level with formula = z2pq/d2 (z = standard normal deviate at 95% confidence interval [CI]; q

= 100 - p), the estimated sample size was 246.[7] If there was no response received or consent could not be obtained from the study participant in the sampled household of the study area, then the next household was considered for the data collection. Furthermore, if onehousehold had two or more geriatric population persons, then a simple random sampling was applied to select the study participant to be included. If only one geriatric personwas staying in the house, and no other family mem-

bers were available, and he was independently able to answer the interview schedule, he was included or else next household was chosen.

The study participants were selected through probability proportional to size sampling from

a line listing of the geriatric population in each village was prepared with the help of Voter's list and Primary sampling units, i.e., the clusters, which were in this case villages, were selected through Probability proportional to Population size method or PPS method. 2 stage cluster sampling was done to select the study participants [Table 1]. The data were collected using a pre-designed, pre-tested, structured schedule for the following:

Outcome variable

Functional status was assessed in terms of:

ADL: Refers to people's daily self-care activities. There are six domains of function: bathing, dressing, toileting, transferring, continence and feeding, adapted from Katz index questionnaire.[8] Cronbach's alpha for the Bengali version was 0.91

IADL: Activities that are not necessary for fundamental functioning, but they let an individual live independently in a community, comprises six domains: food preparation (female)/ shopping (male), housekeeping, laundry, the mode of transportation, the responsibility of own medications of the individual and the ability to handle finances, adapted from Lawton IADL Scale.[9] Cronbach's alpha for the Bengali version was 0.87.

Independent variables

Sociodemographic characteristics : age, sex, religion, caste, literacy, marital status, number of having children, economic status, and type of family were considered. The elderly people who are females or who belong to lower social economic status, who are illiterates are more likely to be neglected in aspects such as optimal health care, nutrition, and healthcareutilization in most of the aspects; this leads to chronic illness which resultsin increased dependence for ADL.

Self-reported morbidity profile (chronic diseases or any illness for 3 months as self-reported by participants.)

Depression: assessed using questions used to elicit the pattern of depressive symptoms in a study by Tamakoshi and Ohno.[10] Cronbach's alpha for the Bengali versionwas 0.73.

Face and content validity of the instrument was actually checked. The tool was translated into the local language (Bengali) maintaining semantic equivalence and pretesting of the questionnaire was done among 30 geriatric people residing in a village of Singur,RHUTC of AIIH&PH. The study was conducted after the clearance from the ethical consideration. Every participant had given written informed consent, after explaining to them the purpose

of the study and after ensuring the absolute confidentiality.

Operational definitions

In this study the outcome variable (functional status) was assessed in terms of both ADL and IADL which consists of six domains each. Participants were scored YES/NO for dependence in each of the six functions, if Yes (i.e.) activity needs supervision, direction, personal assistance or total care, they were allocated 1 point in that domain and if No, they were allocated 0 point (no assistance). The score ranks adequacy of performance in e a c h o f the six functions. The consolidated score of 6 indicates the severe functional impairment, were fully dependent on others and the score 0 indicates full function.

Study participants were categorized into two groups : according to their highest level of functioning of the study participants as:

Dependent (scores of 1-6, who needs assistance in atleast one of the activities) and

Independent (score 0, no assistance).

Data analysis

Data were analyzed using the SPSS Inc. Released 2007.SPSS for Windows, Version

16.0. Chicago, Illinois, US. Appropriate descriptive statistics, univariate and multivariate logistic regression analysis was performed to assess Functional status (ADL and IADL) and elicit its predictors with 95% CI ($P < 0.05$).

III. Results

In this study the mean age of the study participants was 71.82 with the range of 60-102. About 42.3% were females. 17.1% were sched-

uled caste. 43.1% of the study participants were illiterates. 43.5% of the study participants were widow/widowers. 73.2% lived in the joint family. About 48.7% of the study participants belonged to class IV socioeconomic status. Mean per capita income was Rs. 1823 among the study participants.

Tables 2 and 3 depicts the pattern of dependency for basic ADL and IADL and its relationship with gender showing significant association between gender with ADL (χ^2 = 22.4; d.f. = 1; P $\leq$ 0.05) and IADL (χ^2 = 20.6; d.f. = 1; P $\leq$ 0.05). The prevalence of dependency (at least one of the activities) for ADL was 32.5% among which 19 (7.7%) participants were highly dependent, i.e., need assistance to pursue all the six components of basic daily ADL, among which most, 18 (94.8%) participants were females. About

59.3% were dependent among which 31.3% were highly dependent for IADL.

Univariate logistic regression showed people aged >70 years, female gender, less than primary level education, widowed/separated, who lived in joint family, poorest percentile of economic status, who were depressed and the study participants suffered from multimorbidity (more than two chronic diseases) had the increased odds of dependency for ADL. In multivariate logit regression, after adjustment with significant variables, age adjusted odds ratio (AOR) (CI) (95% CI) - 20.6 (8-53.1) and depression AOR (CI) - 13 (4.5-37) remained significant. In this test as the Hosmer-Lemeshow statistic (goodness of fit test) became nonsignificant, the model is fit to explain the factors determining dependency for ADL among the study population. These factors explain about 51.8% variation in dependency for ADL (Nagelkerke R2 = 0.518) [Table 4].

For IADL, bivariate analysis showed every variable considered were significant except economic status (poorest), whereas multivariable analysis showed people aged 70 years

5.2 (2.3-11.4), widowed/separated 2.3 (1.1-5.2), less than primary level education 3.4

(1.5-7.7), joint family 2.6 (1.1-5.9), depression 2.8 (1.2-6.4) were significant among the study participants. The model is fit (Hosmer-Lemeshow statistic: nonsignificant) andit explains 53.5% variation in dependency for IADL [Table 5].

Table 1: *Selection of study participants through Probability Proportional to Size (PPS) sampling from selected villages*

S. No.	Village	Total elderly pop. (n)	Selected by PPS (n)
1	Anandnagar	313	71
2	Balitipa	65	15
3	Baghdanga	165	38
4	Diara	183	42
5	Chinnamore	272	61
6	Paltagarh	86	19
Total		**1084**	**246**

Table 2: *Distribution of study participants according to their functional status (n=246)*

Activity	Male, n (%)	Female, n (%)	Total, n (%)
ADL items			
Toilet	1 (0.7)	21 (20.2)	22 (8.9)
Bathing	3 (2.1)	20 (19.2)	23 (9.3)
Feeding	14 (9.9)	27 (26.0)	41 (16.7)
Dressing	19 (13.4)	36 (34.6)	55 (22.4)
Continence	17 (12.0)	30 (28.8)	47 (19.1)
Transfer within home	29 (20.4)	51 (49.0)	80 (32.5)
IADL items			
Food preparation/shopping	42 (29.6)	54 (51.9)	96 (39.0)
Housekeeping	51 (35.9)	54 (51.9)	105 (42.7)
Laundry	51 (35.9)	57 (54.8)	108 (43.9)
Responsibility for own medication	35 (24.6)	51 (49.0)	86 (35.0)
Transport (moving out of home)	51 (35.9)	67 (64.4)	118 (48.0)
Ability to handle finance	55 (38.7)	79 (76.0)	134 (54.5)

ADL: Activities of daily living; IADL: Instrumental activities of daily living. Figures denote participants dependent (needing supervision, direction, personal assistance, or total care) for that activity.

Table 3: *Distribution of study participants by number of dependent activities (n=246)*

(a) Activities of daily living (ADL)

No. of activities	Male, n (%)	Female, n (%)	Total, n (%)
6 (highly dependent)	1 (5.2)	18 (94.8)	19 (7.7)
5	2 (50.0)	2 (50.0)	4 (1.6)
4	7 (53.8)	6 (46.2)	13 (5.3)
3	9 (52.9)	8 (47.1)	17 (6.9)
2	2 (50.0)	2 (50.0)	4 (1.6)
1	8 (34.7)	15 (65.3)	23 (9.3)
0 (independent)	113 (68.0)	53 (32.0)	166 (67.5)
Mean±SD, median (range): 4.91±1.9, 6 (0–6)			χ^2=22.4, df=1, P<0.05

(b) Instrumental activities of daily living (IADL)

No. of activities	Male, n (%)	Female, n (%)	Total, n (%)
6 (highly dependent)	26 (33.7)	51 (66.3)	77 (31.3)
5	11 (78.6)	3 (21.4)	14 (5.7)
4	9 (100)	0 (0.0)	9 (3.7)
3	8 (72.7)	3 (27.3)	11 (4.5)
2	1 (9.1)	10 (90.9)	11 (4.5)
1	12 (50.0)	12 (50.0)	24 (9.3)
0 (independent)	75 (75.0)	25 (25.0)	100 (40.7)
Mean±SD, median (range): 3.36±2.6, 5 (0–6)			χ^2=20.6, df=1, P<0.05

Table 3: *Multivariable predictors of functional dependency in activities of daily living (ADL) among the elderly (n=246)*

Predictor	Adjusted OR	95% CI
Age >70 years	20.6	8.0–53.1
Depression	13.0	4.5–37.0

The final ADL model had Nagelkerke's R^2=0.518 and a non-significant Hosmer–Lemeshow goodness-of-fit test.

Table 4: *Multivariable predictors of functional dependency in instrumental activities of daily living (IADL) among the elderly (n=246)*

Predictor	Adjusted OR	95% CI
Age >70 years	5.2	2.3–11.4
Widowed/separated	2.3	1.1–5.2
Education <primary level	3.4	1.5–7.7
Joint family	2.6	1.1–5.9
Depression	2.8	1.2–6.4

The final IADL model had Nagelkerke's R^2=0.535 and a non-significant Hosmer–Lemeshow goodness-of-fit test.

IV. Discussion

The present study was conducted to assess the functional status ADL and IADL, which is an index of physical, mental and social well-being of geriatric people. Most of the studies among the elderly concentrate on morbidity and nutritional status, the ability to pursue basic daily activities is of much importanance, yet has been neglected, and very few studies have been conducted in this particular domain or aspect.

Prevalence of dependency

In the present study, dependency (for at least one of the ADL activities) was 32.4%, similar to study in rural Bengaluru (32.4%), whereas studies in rural Haryana (17.6%) and coastal Karnataka (6.5%) showed much less prevalence, it may be due to different study area and instrument used.[11-13] In the present study, among the highly dependent (score 6), most of them were females, whereas in a study in Puducherry showed males were highly dependent.[14] The studies carried out in developed countries such as Malaysia (14.4%), the

USA (15%), and Japan (20%) showed lesser prevalence of ADL disability than in the present study because of their social security system, including institution based unpaid care.[15-17] In the present study, 31.3% were highly dependent (all six activities) for IADL, whereas a study in Shimla showed 21.8% as in our country elderly people handover all the tasks, especially monetary issues, housekeeping totheir sons/daughter.[18] As most of the geriatric people become physically weak, even to move outside the home, they seek help. In the present study, IADL dependency(at least one of the IADL activities component) was 59.3%, whereas studies done in Poland and Norway showed the much less prevalence of at least one problem with IADL reported to be 43.19% and 19.9%, respectively.[19,20] In a study conducted in Spain showed 34.6% were categorized as dependent for at least one ADL and 53.5% for IADL which is similar to the present study, whereas a study conducted on oldest-old Brazilians showed less prevalence (28.7%) poor functional status which is due to the variation in measurement tool offunctional status.[21,22]

Predictors of poor functional status

In the present study and in other studies reviewed, ithas been observed that geriatric people or older age and female gender are important predictors of dependency. Aging leads to physical, mental and social weakening, as a result, they were prone to develop chronic illnesses which in turn lead to dependency and hence, the preventive methods should be provided at an early age, to have a healthy aging.[7,12,13,16-20] Females are still neglected in terms of food and care with a very little focus on their health and well-being, it is due to the gender-segregation related activities in our society where females were restricted to the kitchen work and they are not allowed to get involved in the economic reasoning and decisions. It has been observed that widow/separated were prone to have more dependency in this study and this was also seen in a study in state of Haryana.[12]. Another longitudinal study in India showed household wealth, religion and caste showed no clear association with ADL limitations, similar to this present study.[7]

Poor functional status is significantly associated with the condition of depression and

chronic illnesses. In a study done in Malaysia showed depression (odds ratio [OR] - 1.8) and the presence of >2 chronic illnesses (OR - 8.4), whereas in the present study, there were higher odds of 12 and 6.6, respectively, as significant predictors of debilitating functional status.[15] Studies done in Brazil and Poland showed that depressed and multi-morbid persons having higher odds (OR - 2-3) of d e v e l o p i n g poor functional status.[20,21] In a longitudinal study done in America showed the development of high

depressive symptoms predicted with the greater risk of functional disability, explained 58.2% of the variation in risk factors, which was broadly comparable with the present study.[24] Chronic illness impairs the functional status and reduce physical activity, in turn, leads which may further reduce physical activity and contribute to a cycle of disability. In this study, older age and depression became to be imperative predictors after adjusting with all other variables of ADL. In addition to these, literacy and marital status were the other significant factors affecting IADL.

Strength

Most similar studies were conducted in clinical settings, whereas the current community-based study used a two-stage cluster sampling technique to select participants from 64 villages.

Limitation

However, other factors may also contribute to poor functional status. Prospective studies may provide a clearer picture; additionally, self-reported measures may have introduced subjective bias.

V. Conclusion

The study concluded that 32.5% and 59.3% were dependent for basic ADL and IADL respectively with geriatric people and depression the important predictorsof dependency for ADL. The functional disability among the elderly has become an important public health problem, has provided the impact t h a t they have it on their quality of life of individual, the family, and the health services. More focused emphasis should be given on the integration of geriatric care at the primary care level along with the other levels. There should be a provision for community-based comprehensive geriatric health assessment, as it enables geriatric people to avert illness at early stage, to delay the onset of disabling diseases and conditions and to provide basic domiciliary care and rehabilitation and specialist services at facility level. Social security measures like old age pension should be arranged more liberally and with a great aim for the aged people, especially for the disabled. This will ensure their active participation in the upliftment of the society and they thus will be less dependent on their progeny and will lead a healthy life.

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Case Study / Case Report

Management of Mesiodens: A Supernumerary Tooth

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Abstract
Supernumerary tooth is a developmental disturbance associated with the number of teeth. Mesiodens is a supernumerary tooth present in the midline between the two central incisors. It usually results in oral problems such as malocclusion, food impaction, poor aesthetics, and at times cyst formation. This case report deals with the management of a palatally placed, fully erupted mesiodens in the maxillary arch of a 7-year-old male patient, extracted under local anaesthesia following radiographic confirmation of complete root formation.

Key Words: Dental anomalies, Mesiodens, Pediatric dentistry, Supernumerary teeth, Surgical extraction

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I. Introduction

Supernumerary teeth or hyperdontia refers to the presence of excessive teeth than the normal count in the dental arch. The prevalence of supernumerary teeth is 2.2% in the anterior region among which 78.4% represents mesiodens[1]. It is usually found in primary dentition.[2] Mesiodens is often reported with the complications such as spacing, crowding, speech difficulties, resorption of adjacent root, poor esthetics, delayed eruption of permanent incisors, and development of a dentigerous cyst[1]. The anomaly is usually noticed around 7 to 9 years, and most mesiodens are found during the mixed dentition period. Timely removal of mesiodens leads to normal eruption of the central incisors, owing to the eruption path of the lateral incisors and canine.[2] Extraction is generally the preferred treatment even for erupted mesiodens.[3] That is why mesiodens are seldom observed in adults with permanentdentition as most of them get extracted within the first or early second decade of life.[2]

II. Case Report

This report describes the diagnosis and management of a fully erupted mesiodens in a pediatric dental patient.

CASE REPORT

A seven-year-old male patient reported to the Department of Dentistry with the chief complaint of forwardly placed upper anterior teeth and the presence of an extra tooth behind these two teeth. No significant history of presenting illness was revealed. Family history revealed that his mother underwent mesiodens extraction when she was eight years old. Patient also revealed that he was unhappy with the appearance of his teeth. On examination a palatally placed mesiodens was found in relation to 11, 21 (Figure 1). On radiographic examination (Figure 2), a solitarymesiodens with complete root formation was observed. The mesiodens was just pierced the overlying soft tissue mucosa and was erupting into the oral cavity. Extraction ofmesiodens w planned. Treatment plan was discussed wi his father and written informedconsent was taken. The periosteum was elevated using a periosteal elevator, local anesthesia was then administered and the mesiodens was extracted using tooth forceps (Figure 3, 4). The extraction socket was then irrigated thoroughly with 5

III. Discussion

Mesiodens is the commonly encounteredsupernumerary tooth in dental clinics.1Nearly two-thirds of cases of mesiodens occur in males with a mean age of diagnosis being 6.35±1.85 years.[2] Mostly it is diagnosed during the first decade of life.It'sdevelopment begins during the third trimester ofpregnancy.[1] Mesiodens most often have clinical implications such as delayed eruption, rotation or root resorption of adjacentteeth, malocclusion, and midline diastema. At times, theymay be associated with cystic lesions or even erupt in thenasal cavity.[3] Only less than 20% of cases of mesiodensare free of symptoms without any associated pathology.In the asymptomatic cases of mesiodens, the anomalyis usually diagnosed during routine clinical/radiographic examination.[4],5Yet, in this particular case, the patient did not have any complaints or prior visits to a dental clinic; therefore, the tooth remained unnoticed.

Etiology

There are numerous theories put forwarded byvarious authors to explain the etiology of mesiodens.[6]

Dichotomy theory: Due to certain environmentalfactors like trauma, inflammation, abnormalpressure the tooth germ is divided into two equalor unequal sections resulting in the formation ofsupernumerary teeth.

Genetic theory: Considering the familial pattern,a genetic basis has been suggested for supernumeraryteeth. Nonsyndromic mesiodens is associatedwith the interaction between BMP, SHH, WNT signaling pathways.[7] Decreased expression of USAG and ACVR2A (Activin type II receptor protein) further activates BMP-induced odontogenesis of the supernumerary tooth bud.[8,9,10] Increased WNT and BMP signaling has been reported to produce supernumerary teeth in mice.[11]

Hyperactivity theory: This is the most accepted theory. According to this theory, remnants of the dental lamina develop into an extra tooth bud, resulting in supernumerary tooth.[1]

Associated syndromes: Mesiodens is associatedwith syndromes such as cleidocranial dysplasia (CCD), Gardner's syndrome, cleft palate, Nance-Horan syndrome, Down syndrome, tricho-rhino-phalangeal syndrome, Rothmund-Thomson syndrome.1In the present case, the mesiodens was not associated with any symptoms.

Classification

Mesiodens can be classified based on its shape as Eumorphic or Dysmorphic: conical, tuberculate, molariform, and odontome types.[12]

Recently, a new classification system was given by Albert A and Mupparapu M (2018) for better treatment planning. The classification is based on the angulation of mesiodens to the normal eruptive pattern within the maxilla.[13]

Class 1: Parallel/ 00 to normal eruptive pattern

Class 2: 0-900 to normal eruptive pattern

Class 3: Perpendicular/ 900 to normal eruptive pattern

Class 4: 90-1800 to normal eruptive pattern

Class 5:Inverted/ 1800 to normal eruptive pattern

Mesiodens can be diagnosed by the combination of both clinical and radiographic examination. Abnormal midline diastema in early mixeddentition or proclination of teeth gives a clue to the presence of mesiodens. Radiographic examination shows a tooth-like structure in between the incisors. Generally, Clark's rule (also known as the SLOB technique – same lingual, opposite buccal) is followed to confirm the precise location of the mesiodens.[1]

Different timing for the extraction is suggested in the literature: early intervention and delayed intervention.[14,15] Early intervention, extraction immediately its after diagnosis, is the most preferred. Extraction of the mesiodens in the early mixed dentition period promotes spontaneous eruption and alignment of central incisors. Also, it promotes the early orthodontic intervention there by improving the patient profile at an early age. Late intervention refers to the extraction after the complete root formation of adjacent teeth. This is suggested to avoid potential complications to the developing roots of adjacent teeth.[1]

In the present case the root was completely formed, the crown was easily visible, and the positioning was favourable, so extraction with forceps was planned. The patient was scheduled for three-monthly follow-up for self-correction of the proclination of the upper central incisors, otherwise orthodontic minor correction was planned.

IV. Conclusion

Clinical experience suggests and the dental literature supports that an individualized radiographic examination of any pediatric patient who presents clinical evidence of delayed permanent tooth eruption or temporary tooth displacement with or without a history of previous dental trauma should be performed. Early diagnosis allows adoption of a less complex and less expensive treatment and ensures better prognosis.

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Case Study / Case Report

Scrub Typhus Presenting as Severe Pneumonia and Acute Febrile Illness in a Malnourished Infant with Infantile Tremor Syndrome

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Abstract

Background: Scrub typhus is an important cause of acute undifferentiated febrile illness in endemic regions of India. Clinical manifestations are often nonspecific and may mimic pneumonia, meningitis, Kawasaki disease, or sepsis, making early diagnosis difficult. **Clinical Description:** A 6-month-old male infant presented with fever for 10 days, cough and respiratory distress for 4 days, and abdominal distension. The child had severe wasting and was underweight, with features suggestive of infantile tremor syndrome, including hyperpigmentation of the knuckles and tongue fasciculations. Examination revealed tachypnea, bilateral conjunctival congestion, cervical lymphadenopathy, hepatomegaly, and hypoxemia. Investigations showed anemia, thrombocytopenia, transaminitis, hyponatremia, low vitamin B12 levels, and left middle zone opacity on chest radiograph. Ultrasonography demonstrated ascites. Scrub typhus IgM was reactive while dengue serology was negative. **Management and Outcome:** The child was initially treated with empirical intravenous antibiotics; doxycycline and parenteral vitamin B12 were subsequently started after serological confirmation. Fever subsided within 48 hours with marked clinical improvement. **Conclusion:** Scrub typhus should be considered in infants presenting with prolonged fever, respiratory symptoms, hepatomegaly, thrombocytopenia, and transaminitis in endemic areas. Early diagnosis and prompt doxycycline therapy can be lifesaving. Coexisting malnutrition and infantile tremor syndrome may further obscure the diagnosis.

Key Words: Infantile tremor syndrome, Pediatric infectious disease, Prolonged fever, Scrub typhus, Severe pneumonia

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I. Introduction

Scrub typhus is a zoonotic infection caused by Orientiatsutsugamushi and transmitted through the bite of infected larval trombiculid mites (chiggers). It is increasingly recognized as a major cause of acute undifferentiated febrile illness in the "tsutsugamushi triangle," including several regions of India. Clinical manifestations range from mild febrile illness to severe multiorgan dysfunction involving the respiratory, neurological, hepatic, and cardiovascular systems.

In children, diagnosis is particularly challenging because early symptoms overlap with common tropical infections such as dengue, malaria, enteric fever, viral illnesses, and bacterial pneumonia. Presence of malnutrition or associated nutritional deficiencies may further complicate the clinical picture.

We report a case of scrub typhus in a 6-month-old infant presenting predominantly with severe pneumonia and features of infantile tremor syndrome, highlighting the diagnostic dilemma and importance of early suspicion in endemic regions.

II. Clinical Description

A 6-month-old male infant was admitted with fever for 10 days, cough and shortness of breath for 4 days, and abdominal distension for 4 days.

The child had a history of two previous hospitalizations for cough and respiratory distress, during which he received intravenous antibiotics for 5-7 days. He was born at term by normal vaginal delivery with a birth weight of 2.8 kg and was immunized appropriately for age. Feeding history revealed breastfeeding along with diluted cow's milk.

Family history was significant for smoking exposure at home and one neonatal death.

On examination, the infant appeared irritable and toxic-looking with severe wasting and underweight. He had tachypnea, cracked lips, bilateral conjunctival congestion, hyperpigmentation of knuckles, and tongue fasciculations. Oxygen saturation was 92% on room air. Bilateral cervical lymph nodes were palpable. Respiratory examination showed increased work of breathing with bilateral vesicular breath sounds and crepitations over the left hemithorax. Abdomen examination revealed hepatomegaly with liver palpable 2.5 cm below the right costal margin. Pallor was present, while icterus and clubbing were absent.

Initial differential diagnoses included severe pneumonia, Kawasaki disease, primary immunodeficiency, scrub typhus, and meningitis.

Management and Outcome:Laboratory investigations revealed hemoglobin 8.3 g/dL, total leukocyte count 18,000/cu mm with lymphocytic predominance, platelet count 1.2 lakh/cu mm, elevated transaminases (SGOT/SGPT: 82/46 IU/L), hyponatremia (Na+ 127 mEq/L), and low serum vitamin B12 levels. Chest radiograph demonstrated left middle zone opacity suggestive of pneumonia. Ultrasonography of abdomen revealed free fluid in the pelvis and hepatorenal pouch with contracted gall bladder. Dengue IgM was nonreactive, whereas scrub typhus IgM was reactive. Cerebrospinal fluid examination was unremarkable.

The child was initially managed with oxygen supplementation, nebulization, and empirical intravenous ceftriaxone and amikacin. Following confirmation of scrub typhus serology, doxycycline therapy was initiated along with parenteral vitamin B12 supplementation.

Clinical improvement was remarkable. Fever subsided within 48 hours of starting doxycycline, abdominal distension gradually decreased, respiratory distress improved, and the child became playful with reduced irritability.

The final diagnosis was scrub typhus with infantile tremor syndrome.

III. Discussion

Scrub typhus has emerged as a significant re-emerging pediatric infection in India. The disease often presents with nonspecific symptoms, resulting in delayed diagnosis and treatment. Classical eschar may be absent in many pediatric patients, particularly infants, thereby increasing diagnostic difficulty.

Our patient presented predominantly with prolonged fever, respiratory distress, hepatomegaly, thrombocytopenia, transaminitis, and hyponatremia, all of which are recognized manifestations of scrub typhus. Pulmonary involvement in scrub typhus may range from mild pneumonitis to severe pneumonia and acute respiratory distress syndrome.

The coexistence of severe malnutrition and infantile tremor syndrome further complicated the presentation. Features such as hyperpigmentation of knuckles, tongue fasciculations, macrocytosis, and low vitamin B12 levels supported the diagnosis of infantile tremor syndrome. Nutritional deficiencies may impair immunity and predispose to severe infections.

An important clinical clue in this case was the dramatic defervescence within 48 hours of initiating doxycycline, which strongly supported the diagnosis of scrub typhus. Early initiation of appropriate therapy significantly reduces morbidity and mortality.

This case emphasizes that scrub typhus should remain an important differential diagnosis in children presenting with prolonged fever and respiratory symptoms in endemic areas, especially when associated with thrombocytopenia, transaminitis, or hepatomegaly.

Learning Points

- Scrub typhus should be considered in prolonged febrile illness with respiratory distress in endemic regions.
- Thrombocytopenia, transaminitis, hyponatremia, and hepatomegaly are important diagnostic clues.
- Absence of eschar does not exclude scrub typhus.
- Early doxycycline therapy produces rapid clinical improvement.
- Coexisting malnutrition and vitamin B12 deficiency may complicate presentation and management.

Consent

Written informed consent was obtained from the child's guardians for publication of this case report and clinical details.

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Review Article

Vestibulo-Cerebellar Modulation of Stress-Related Immune Responses: A Narrative Review

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Abstract

The vestibulo-cerebellum occupies a distinctive anatomical position, with connections to the hypothalamic-pituitary-adrenal (HPA) axis, the sympathetic-adrenomedullary (SAM) system, and the central noradrenergic system centred on the locus coeruleus. Because immune cells and lymphoid organs carry receptors for stress hormones and noradrenergic innervation, changes in these neuroendocrine pathways during stress can, in principle, alter immune function. A body of largely older, predominantly animal-based experimental work has examined whether vestibular and vestibulo-cerebellar stimulation can modulate stress-induced immune alterations, and this narrative review summarises that literature. We describe the physiology of the HPA and SAM stress axes and their known effects on immune cells; the neuroanatomical connections linking the vestibulo-cerebellum to the locus coeruleus and limbic-hypothalamic structures; proposed mechanisms of vestibular stimulation, including GABAergic modulation of anterior pituitary secretion and hippocampal involvement; and the preclinical and limited clinical evidence for vestibular neuro-immunomodulation. Vestibular stimulation is not a single, uniform intervention: rotational, galvanic, and caloric paradigms differ physiologically, and rotational stimulation in particular co-activates multiple sensory and autonomic pathways beyond the vestibulo-cerebellum itself, which complicates attribution of any observed immune effect to the vestibulo-cerebellum specifically. The available evidence is not sufficient to establish that vestibular stimulation reproducibly and specifically modulates immunity through the vestibulo-cerebellum in humans. We conclude that vestibular and vestibulo-cerebellar pathways may participate in the regulation of stress-related immune responses through interactions with neuroendocrine and autonomic systems, but that the precise neural circuitry, the relative contribution of individual pathways, and the translational relevance to humans remain incompletely defined, warranting well-designed experimental and clinical studies.

Key Words: HPA axis, Neuroimmunomodulation, Stress, Vestibular stimulation, Vestibulo-cerebellum

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I. Introduction: Stress Physiology and Immune Modulation

Since the time of Galen, it has been observed that mental state is linked to physical disease susceptibility[1]. A substantial literature now links psychosomatic and immune dysfunction to the body's physiological response to stress, mediated through homeostatic regulatory mechanisms whose chronic imbalance is reflected in somato-visceral dysfunction and disordered cognitive function[1].

Psychological stress is the neurological response to environmental demands that exceed an individual's coping capacity. The relationship between stress and immunity is more nuanced than a simple "stress suppresses immunity" framing suggests. A meta-analysis of three decades of inquiry found that acute stress can produce potentially adaptive changes in innate immunity — including lymphocytosis, mobilisation of cytotoxic lymphocyte subsets such as NK cells and CD8+ T cells, and increased NK-cell cytotoxicity — while chronic stress is more consistently associated with suppression of several immune measures[2]. More recent work similarly frames the neuroimmune response not as a unidirectional suppression pathway, but as an integrated response involving neural, endocrine, and immune mediators that can produce either activation/inflammation or suppression depending on the nature, duration, and context of the stressor[3].

Two neuroendocrine axes principally mediate this response. The hypothalamic-pituitary-adrenal (HPA) axis proceeds: hypothalamus $\rightarrow$ corticotropin-releasing hormone (CRH) $\rightarrow$ pituitary adrenocorticotropic hormone (ACTH) $\rightarrow$ adrenal glucocorticoids $\rightarrow$ immune cells. The sympathetic-adrenomedullary (SAM) system proceeds: sympathetic nervous system $\rightarrow$ noradrenaline/adrenaline $\rightarrow$ β-adrenergic receptors on immune cells. Table 1 summarises how the pattern of neuroendocrine activation differs by stress type and the immune consequences typically associated with each.

Stress type	**Typical neuroendocrine response**	**Potential immune consequence**
Acute stress	SNS/HPA activation	Redistribution/activation of selected immune cells
Repeated/short-term stress	Persistent neuroendocrine activation	Variable immune effects
Chronic stress	Prolonged HPA/SNS dysregulation	Immune suppression and/or chronic inflammation

Table 1: *Neuroendocrine and immune consequences by stress type (reproduced from reviewer feedback)*

Human evidence for stress-related immune modulation includes vaccine-response studies: medical students under examination stress showed reduced antibody and T-cell responses to hepatitis B vaccination, and caregivers under chronic stress produced lower IgG antibody titres to bacterial and meningococcal conjugate vaccines than controls, with social support identified as an ameliorating factor[3]. Chronic stress has also been associated with increased production of pro-inflammatory cytokines [3] which over prolonged inflammation poses a tissue-damage risk.

II. Neuroanatomy of the Vestibulo-Cerebellum and Its Connections

The vestibular nuclei send both excitatory and inhibitory projections to multiple brain regions; derangement of these circuits can alter sensorimotor behaviour. The vestibulo-cerebellum is anatomically positioned with links to the limbic-hypothalamic-striatal-diencephalic region and midbrain-reticular-vestibulomedullary structures, and in particular to the central noradrenergic system centred on the locus coeruleus[5,6]. Through these connections it is proposed to act as a neural switch mechanism integrating somato-visceral responses during stress, receiving signals conveyed by multiple neurotransmitters, peptides, and hormones[6].

Previous experimental studies have reported that lesions of the vestibulo-cerebellum produce measurable effects on immune function, including modulation of hypersensitivity-linked immune responses and humoral immunity, in animal models[7,8]. These findings should be read as animal, lesion-based experimental evidence rather than established human physiology; they motivate the mechanistic hypotheses discussed below without themselves demonstrating a clinically relevant effect in humans.

III. Mechanisms of Vestibular Stimulation

Vestibular stimulation is not a single, uniform intervention. Rotational stimulation, galvanic vestibular stimulation, caloric stimulation, and other vestibular paradigms differ in the receptors and pathways they engage and are understood to have distinct physiological effects; findings from one paradigm should not be generalised to the others without direct evidence.

Rotational stimulation at low frequency has been proposed as a non-invasive means of activating the vestibulo-cerebellum via the vestibular nuclei[6]. However, deterministic rotational motion — with its acceleration, deceleration, and abrupt-stoppage components — co-activates a wide range of additional receptors beyond the semicircular canals: the retina, ocular muscles, proprioceptors of the neck, face, body, and trunk, joint receptors, visceral interoceptors, and baroreceptors. This is an important conceptual limitation of the literature discussed here: **vestibular stimulation is not equivalent to selective vestibulo-cerebellar activation**, and rotational paradigms in particular cannot cleanly isolate a vestibulo-cerebellar-specific effect from the contribution of these co-activated sensory and autonomic pathways. Where the literature below reports an immune effect of "vestibular stimulation," this caveat should be assumed throughout. Specific stimulation parameters — frequency in Hz, angular velocity, acceleration/deceleration profile, and exposure duration — are not consistently reported in the sources reviewed here and are not reproduced in this review for that reason.

Two candidate mechanistic pathways have been proposed. First, vestibular stimulation has been reported to increase GABA release, which has separately been shown to modulate anterior pituitary hormone secretion [9]; whether this GABAergic pathway is the operative link between vestibular stimulation and HPA-axis inhibition specifically — as opposed to a parallel, unrelated observation — is not established by the cited evidence, and the underlying studies do not report whether the effect was blocked by GABA-receptor antagonism. Second, vestibular stimulation has been proposed to activate the hippocampal f tion, which in turn inhibits the HPA axis [10] Both pathways should be regarded as hypotheses requiring direct confirmation rather than established mechanisms.

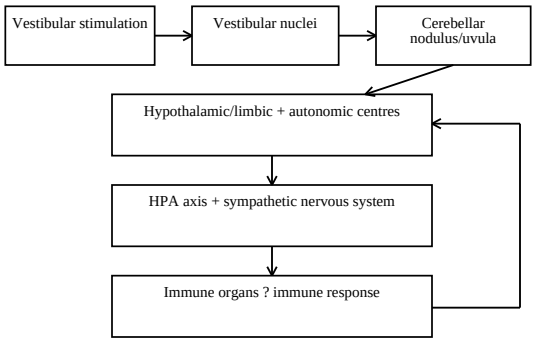
IV. Preclinical and Clinical Evidence of Vestibular Neuro-Immunomodulation

Neuroimmunomodulation is understood to involve close interaction between the endocrine, nervous, and immune systems. Several hormones and peptides — ACTH, growth hormone, endorphins, enkephalins, POMC-related

peptides, neuropeptide Y, and interleukin-1 — are synthesised by cells of both the neuroendocrine and immune systems, and extensive sympathetic innervation of lymphoid organs, with nerve endings and varicosities in close contact with immune cells, provides a plausible hard-wired route for neuronal regulation of immune function[11]. Broader signalling within this neuroendocrine-immune network has been described[12], including hor
pituitary cells in response to interleukin-1 13.

Directly relevant experimental work is limited in volume and age. An early rat study reported that restricted low-frequency vestibular stimulation corrected stress-induced alterations in immune status[14,15]. This finding is supportive of the review's central hypothesis but is a single, decades-old animal experiment; it does not by itself establish clinical efficacy or mechanism in humans, and should not be read as having done so.

A further source is cited for context on neu- odulation in brain disorders more broadly 16.



V. Methodological Note

This is a narrative, not a systematic, review. Sources were drawn from the author's prior publications on this topic and the general stress-neuroimmunology literature, rather than a systematic database search with defined inclusion criteria. Quantitative synthesis (meta-analysis) was not attempted, and the review does not claim comprehensive coverage of the field.

VI. Conclusion

Available experimental evidence suggests that vestibular and vestibulo-cerebellar pathways may participate in the regulation of stress-related immune responses through interactions with neuroendocrine and autonomic systems. However, the precise neural circuitry, the relative contribution of individual neuroendocrine pathways, and the translational relevance to humans remain incompletely defined. Well-designed experimental and clinical studies — particularly ones that isolate vestibulo-cerebellar activation from the other sensory and autonomic pathways co-activated by rotational stimulation — are required to determine whether vestibular stimulation has reproducible and clinically meaningful immunomodulatory effects.

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AUTHOR GUIDELINES

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10. Acknowledgement, if any. One or more statements should specify: (1) contributions that need acknowledging but do not justify authorship, such as general support by a departmental chair; (2) acknowledgments of technical help; and (3) acknowledgments of financial and material support, which should specify the nature of the support. This should be included on the title page of the manuscript, not in the main article.
11. Conflict of interest: if any.

Abstract Page

The abstract page should contain: the type of article; the title of the article; the abstract; and key words.

The abstract may be structured or unstructured, per the category limits in the table below. It should be informative and completely self-explanatory, briefly present the topic, state the scope of the experiments, indicate significant data and point out major findings and conclusions. Standard nomenclature should be used and abbreviations should be avoided. No literature should be cited in the abstract.

Key Words: following the abstract, list not more than 6 key words, in alphabetical order, to support indexing.

Text Page

The article should be written in the following format:

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This should be brief and the review of the literature should be pertinent to the theme of the paper. Extensive review and unnecessary detail of earlier work should be avoided. State the purpose and summarize the rationale for the study or observation.

Materials and Methods

This should cover all relevant aspects: type of study, study design, study population, period of study, study area, tools and techniques used and statistical analysis, with Ethics Committee approval stated.

Results

Present results in a logical sequence in the text, tables and illustrations, giving the main or most important findings first. Do not repeat in the text all the data in the tables or illustrations; emphasize or summarize only important observations.

Discussion

Include a summary of key findings; results as they relate to a prior hypothesis; comparison with similar work; strengths and limitations of the study; what the study adds to the available evidence; effects on patient care and health policy; and future research directions. Do not repeat in detail data or other material already given in the Introduction or Results.

Conclusion and Recommendations

State the conclusions drawn from the findings, linked to the objectives of the study, with recommendations where appropriate.

References

References should be numbered consecutively in the order in which they are first mentioned in the text (not alphabetical order). Identify references in text, tables and legends by Arabic numerals in superscript after punctuation marks. References cited only in tables or figure legends should be numbered according to the sequence established by the first identification, in the text, of that table or figure. Use the style based on the formats used by the NLM in Index Medicus. Information from manuscripts submitted but not accepted/published should be cited in the text as unpublished observations, with written permission from the source.

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JMN Journal of Medical Sciences

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JMN Journal of Medical Sciences is an official publication of JMN Medical College and Hospital, which started its journey in 2023 with the MBBS course. The journal is a double-blind peer-reviewed, open-access journal with a frequency of publication of twice a year (biannual).

Aim of the Journal

JMN Journal of Medical Sciences aims to disseminate reliable, evidence-based knowledge in the field of medicine and health care. It provides a platform for publishing research findings, newer developments and knowledge in medical sciences to all faculty members, postgraduate trainees and undergraduate students. The journal helps health care professionals stay updated with new discoveries, diagnostic methods, treatments and patient care services. It promotes critical thinking, scientific discussion and exchange of thoughts among researchers and medical practitioners, thus improving and supporting evidence-based practice and contributing to better patient care and health outcomes.

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